

# Strong local convergence of mollified derivatives

MathlibAnnex.Mollification.tendsto\_eLpNorm\_fderiv\_mollify\_sub

## theorem

Approximates the derivative in local  $L^n$ , where  $n$  is the dimension of the domain.

## Statement

Fix a positive integer  $n$  and an integer  $N \geq 0$ . Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}^N$  and  $C \geq 0$  satisfy

$$\|f(x) - f(y)\|_\infty \leq C\|x - y\|_\infty \quad (x, y \in \mathbb{R}^n),$$

and let  $K \subseteq \mathbb{R}^n$  be compact. For  $\varepsilon > 0$ , let  $\rho_\varepsilon$  be the chosen smooth nonnegative bump kernel centered at zero, with inner radius  $\varepsilon/2$  and outer radius  $\varepsilon$ , divided by its positive Lebesgue integral. Thus  $\int_{\mathbb{R}^n} \rho_\varepsilon(y) dy = 1$  and its topological support is contained in  $\{y : \|y\|_\infty \leq \varepsilon\}$ . Using this same kernel, define the normalized convolution

$$f_\varepsilon(x) = \int_{\mathbb{R}^n} \rho_\varepsilon(y) f(x - y) dy.$$

Then

$$\|Df_\varepsilon - Df\|_{L^n(K)} = \left( \int_K \|Df_\varepsilon(x) - Df(x)\|_{\text{op}}^n dx \right)^{1/n} \longrightarrow 0 \quad (\varepsilon \downarrow 0),$$

and  $Df_\varepsilon \in L^n(K)$  for all sufficiently small positive  $\varepsilon$ . Here  $\|\cdot\|_{\text{op}}$  is induced by the two sup norms. The exponent is the already fixed domain dimension, not another parameter.

## Assumptions

The dimension satisfies  $n \geq 1$ , while  $N$  may be zero. The compact set  $K$  may be empty or have empty interior. The displayed Lipschitz bound holds on the whole domain. Neither  $f$  nor  $Df$  is assumed to belong to  $L^n$  on the whole space. The derivative is defined almost everywhere by Rademacher's theorem; choosing zero at nondifferentiability points gives the same  $L^n$  quantities.

## Conclusion

Both the seminorm convergence and eventual  $L^n$  membership hold for Lebesgue measure restricted to  $K$ . The original derivative itself belongs to  $L^n(K)$  by measurability and the bound  $\|Df\| \leq C$ .

## Notes

Linearity of mollification for locally integrable summands and a common compact support bound for mollifications of a compactly supported map will be used later. Those auxiliary results concern the same kernels and are cited here; the present theorem does not assume that  $f$  has compact support.

## Proof route

First prove  $Df_\varepsilon = \rho_\varepsilon * Df$ . On  $K$ , each coordinate of this convolution equals the convolution of a bounded, compactly supported scalar function. Scalar convergence and a finite coordinate estimate then give the required operator-norm convergence.

### ▼ Proof steps

1. **Differentiate the convolution.** Fix  $\varepsilon > 0$  and  $x \in \mathbb{R}^n$ . Rademacher's theorem and preservation of Lebesgue-null sets by  $y \mapsto x - y$  give differentiability of  $f$  at  $x - y$  for almost every  $y$ . For every increment  $h$ ,

$$\|\rho_\varepsilon(y)[f(x + h - y) - f(x - y)]\|_\infty \leq C\rho_\varepsilon(y)\|h\|_\infty, \quad \int C\rho_\varepsilon(y) dy = C.$$

The integrand is integrable for each  $x$ , because  $f$  is continuous and the kernel has compact support. The proposed derivative is measurable and is bounded by the same integrable function. These are the hypotheses for differentiation under the integral, which gives

$$Df_\varepsilon(x) = \int_{\mathbb{R}^n} \rho_\varepsilon(y) Df(x - y) dy.$$

For the  $a$ th input vector  $e_a$  and the  $b$ th output coordinate, this reads

$$(Df_\varepsilon(x)e_a)_b = \int_{\mathbb{R}^n} \rho_\varepsilon(y) (Df(x - y)e_a)_b dy.$$

2. **Replace a derivative coordinate by a compactly supported one without changing its convolution on  $K$ .** Choose  $R \geq 0$  with  $K \subseteq \overline{B}_\infty(0, R)$  and put  $S = \overline{B}_\infty(0, R + 1)$ . For  $1 \leq a \leq n$  and  $1 \leq b \leq N$ , define the scalar function

$$q_{ab}(z) = \begin{cases} (Df(z)e_a)_b, & z \in S, \\ 0, & z \notin S. \end{cases}$$

It is measurable,  $|q_{ab}(z)| \leq C$ , and it vanishes outside the compact set  $S$ . Hence  $q_{ab} \in L^n(\mathbb{R}^n)$ .

If  $x \in K$ ,  $0 < \varepsilon < 1$ , and  $\rho_\varepsilon(y) \neq 0$ , then

$$\|x - y\|_\infty \leq \|x\|_\infty + \|y\|_\infty < R + 1, \quad x - y \in S.$$

Consequently, for every  $y$ ,

$$\rho_\varepsilon(y)(Df(x - y)e_a)_b = \rho_\varepsilon(y)q_{ab}(x - y).$$

If the kernel factor is nonzero, this follows from  $x - y \in S$ ; otherwise both sides vanish. Also  $q_{ab}(x) = (Df(x)e_a)_b$ . Substituting into Step 1 gives

$$((Df_\varepsilon(x) - Df(x))e_a)_b = (\rho_\varepsilon * q_{ab})(x) - q_{ab}(x) \quad (x \in K, 0 < \varepsilon < 1).$$

3. **Prove convergence for that scalar function.** The normalized bumps have outer radius  $\varepsilon \rightarrow 0$  and outer-to-inner radius ratio 2. The Lebesgue-differentiation convolution theorem therefore gives

$$(\rho_\varepsilon * q_{ab})(x) \longrightarrow q_{ab}(x) \quad \text{for almost every } x.$$

Normalization and nonnegativity give

$$|(\rho_\varepsilon * q_{ab})(x)| \leq \int \rho_\varepsilon(y) |q_{ab}(x-y)| dy \leq C.$$

Set  $T = \overline{B}_\infty(0, R+2)$ . If  $x \notin T$  and  $\rho_\varepsilon(y) \neq 0$  with  $\varepsilon < 1$ , then

$$\|x-y\|_\infty \geq \|x\|_\infty - \|y\|_\infty > R+1,$$

so  $q_{ab}(x-y) = q_{ab}(x) = 0$ . Consequently

$$|(\rho_\varepsilon * q_{ab})(x) - q_{ab}(x)|^n \leq (2C)^n \mathbf{1}_T(x), \quad \int (2C)^n \mathbf{1}_T(x) dx = (2C)^n |T| < \infty.$$

Dominated convergence, followed by taking the  $n$ th root, proves

$$\|\rho_\varepsilon * q_{ab} - q_{ab}\|_{L^n(\mathbb{R}^n)} \longrightarrow 0.$$

Its  $L^n(K)$  norm is no larger. Step 2 therefore proves convergence of every derivative-coordinate difference on  $K$ .

4. **Pass from coordinates to the operator norm.** For a linear map  $A : \mathbb{R}^n \rightarrow \mathbb{R}^N$  and  $\|v\|_\infty \leq 1$ ,

$$\begin{aligned} (Av)_b &= \sum_{a=1}^n v_a (Ae_a)_b, \\ |(Av)_b| &\leq \sum_{a=1}^n |(Ae_a)_b| \leq \sum_{a=1}^n \sum_{b'=1}^N |(Ae_a)_{b'}|. \end{aligned}$$

Taking the maximum over  $b$  and then the supremum over  $v$  gives  $\|A\|_{\text{op}} \leq \sum_{a,b} |(Ae_a)_b|$ . To justify the finite-sum  $L^n$  estimate, fix  $0 < \varepsilon < 1$  and write, on  $K$ ,

$$g_{ab}(x) := ((Df_\varepsilon(x) - Df(x))e_a)_b, \quad h(x) := \sum_{a=1}^n \sum_{b=1}^N |g_{ab}(x)|.$$

Steps 2–3 give measurable  $g_{ab}$  with  $|g_{ab}| \leq 2C$ . Thus  $h \leq 2CnN$  and  $\int_K h^n < \infty$ , because  $K$  has finite measure. For  $n > 1$ , apply Hölder to  $|g_{ab}|$  and  $h^{n-1}$ , with conjugate exponents  $n$  and  $n/(n-1)$ :

$$\begin{aligned} \int_K h(x)^n dx &= \sum_{a,b} \int_K |g_{ab}(x)| h(x)^{n-1} dx \\ &\leq \sum_{a,b} \left( \int_K |g_{ab}(x)|^n dx \right)^{1/n} \left( \int_K h(x)^n dx \right)^{(n-1)/n}. \end{aligned}$$

If  $\int_K h^n > 0$ , divide by its  $(n - 1)/n$  power; if it is zero, the desired inequality is immediate. For  $n = 1$  the same inequality is the equality  $\int_K h = \sum_{a,b} \int_K |g_{ab}|$ . This proves the finite-sum form of Minkowski used here, and hence

$$\begin{aligned} \|Df_\varepsilon - Df\|_{L^n(K)} &\leq \|h\|_{L^n(K)} \\ &\leq \sum_{a=1}^n \sum_{b=1}^N \|x \mapsto ((Df_\varepsilon(x) - Df(x))e_a)_b\|_{L^n(K)} \longrightarrow 0. \end{aligned}$$

The last limit uses Step 3 for each of the finitely many pairs  $(a, b)$ . For  $N = 0$  every operator is zero and the conclusion is immediate. Finally,  $f_\varepsilon$  is smooth for every  $\varepsilon > 0$ . Its derivative is continuous and bounded on compact  $K$ , so  $Df_\varepsilon \in L^n(K)$ . This supplies the membership assertion as well as the convergence.

## Main citations

- Exact declaration and its source — `MathlibAnnex.Mollification.tendsto_eLpNorm_fderiv_mollify_sub`
- Convolution with the normalized kernel — `MathlibAnnex.Mollification.mollify`
- Linearity of convolution for locally integrable maps — `MathlibAnnex.Mollification.mollify_add`
- Smoothness of positive-parameter mollification — `MathlibAnnex.Mollification.contDiff_mollify`
- One compact set for small-parameter supports — `MathlibAnnex.Mollification.eventually_tsupport_mollify_subset_compact`
- Eventual derivative membership on a compact set — `MathlibAnnex.Mollification.eventually_mollify_fderiv_memLpOn_compact`
- Lipschitz derivatives in the dimension-power space — `MathlibAnnex.Mollification.lipschitz_fderiv_memLpOn_compact`

## Lean source signature (exact)

```
theorem tendsto_eLpNorm_fderiv_mollify_sub
  {m N : ℕ} {f : (Fin (m + 1) → ℝ) → (Fin N → ℝ)} {C : ℝ≥0} (hf : LipschitzWith C f)
  {K : Set ((Fin (m + 1) → ℝ))} (hK : IsCompact K) :
  (Tendsto (fun ε => eLpNorm
    (fun x => fderiv ℝ (mollify ε f) x - fderiv ℝ f x)
    ((m + 1 : ℕ) : ℝ≥0∞) (volume.restrict K)) (ℕ[>] (0 : ℝ)) (ℕ 0)) ∧
  ∀ f ε in ℕ[>] (0 : ℝ), MemLp
    (fun x => fderiv ℝ (mollify ε f) x) ((m + 1 : ℕ) : ℝ≥0∞) (volume.restrict K)
```

| In the source                                                 | Mathematical meaning                                                                                                                      |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>{m N : ℕ} {f : (Fin (m + 1) → ℝ) → (Fin N → ℝ)}</code>  | The positive domain dimension is $n = m + 1$ and the output dimension is $N$ . The input is $f : \mathbb{R}^n \rightarrow \mathbb{R}^N$ . |
| <code>{C : ℝ≥0} (hf : LipschitzWith C f)</code>               | A nonnegative constant $C$ and the global bound $\ f(x) - f(y)\ _\infty \leq C\ x - y\ _\infty$ for all $x, y$ .                          |
| <code>{K : Set ((Fin (m + 1) → ℝ))} (hK : IsCompact K)</code> | The integration set $K$ is compact, possibly empty.                                                                                       |

| In the source                                                                                                                   | Mathematical meaning                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fderiv ℝ (mollify ε f) x - fderiv ℝ f x</code>                                                                            | The continuous-linear-map difference $Df_\varepsilon(x) - Df(x)$ ; its norm is the operator norm induced by the sup norms.                                                                                            |
| <code>((m + 1 : N) : ℝ≥0∞) (volume.restrict K)</code>                                                                           | The exponent is the already fixed domain dimension $n$ , and the measure is Lebesgue measure restricted to $K$ . The $L^n$ seminorm takes values in $[0, \infty]$ .                                                   |
| In the source                                                                                                                   | Mathematical meaning                                                                                                                                                                                                  |
| <code>(<math>\mathcal{N}[\&gt;]</math> (<math>\emptyset : \mathbb{R}</math>)) (<math>\mathcal{N} \emptyset</math>)</code>       | The first output is convergence to zero as $\varepsilon \rightarrow 0$ through positive values: $\ Df_\varepsilon - Df\ _{L^n(K)} \rightarrow 0$ .                                                                    |
| <code><math>\forall^f \varepsilon</math> in <math>\mathcal{N}[\&gt;]</math> (<math>\emptyset : \mathbb{R}</math>), MemLp</code> | The second, conjunctive output holds for all sufficiently small positive $\varepsilon$ : $Df_\varepsilon \in L^n(K)$ . Membership includes a.e. strong measurability and finite seminorm, not just a pointwise bound. |

#### ▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section
open Set MeasureTheory Filter
open scoped Convolution Topology ENNReal NNReal Pointwise
namespace MathlibAnnex
namespace Mollification

```

#### ▼ Exact content identity

Declaration: `MathlibAnnex.Mollification.tendsto_eLpNorm_fderiv_mollify_sub`

Accepted content SHA-256: `2b5f64576a032a3d0f209601db7ec4155cdd19e8f87f625f6a7a1437104445b3`

Accepted source guide SHA-256: `e8dda24a67f2e7c8929b348569786f54bb3adbdbabfff4377049c0fc9c1c6317`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`